FIITJEE

ALL INDIA TEST SERIES

FULL TEST – VIII

JEE (Main)-2021

TEST DATE: 07-03-2021

Time Allotted: 3 Hours **Maximum Marks: 300**

General Instructions:

- The test consists of total 90 questions.
- Each subject (PCM) has 30 questions.
- This question paper contains **Three Parts.**
- **Part-A** is Physics, **Part-B** is Chemistry and **Part-C** is Mathematics.
- Each part has only two sections: **Section-A and Section-B**.
- **Section – A** : Attempt all questions.
- **Section – B :** Do any five questions out of 10 Questions.

  Section-A (01 – 20, 31 – 50, 61 – 80) *contains 60 multiple choice questions which have* ***only one correct answer****. Each question carries* ***+4 marks*** *for correct answer and* ***–1 mark*** *for wrong answer.*

  Section-B (21 – 30, 51 – 60, 81 – 90) *contains 30 Numerical answer type questions with answer XXXXX.XX and each question carries* ***+4 marks*** *for correct answer. There is no negative marking.*

Physics

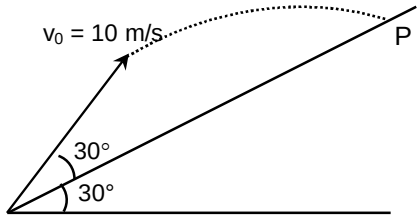
PART – A

SECTION – A
(One Options Correct Type)

This section contains **20 multiple choice questions**. Each question has **four choices** (A), (B), (C) and (D), out of which **ONLY ONE** option is correct.

1. An ideal gas expands according to the law $V = V_0e^{-\alpha p}$. Bulk modulus of gas during the process is (where V is volume, p is pressure and α is a constant)
   (A) $\frac{1}{\alpha}$
   (B) $V_0\alpha$
   (C) $\frac{V_0}{\alpha}$
   (D) α

2. A particle is projected up an inclined plane of inclination 30° with the horizontal as shown in the figure. It strikes the inclined plane at point P. If magnitude of the initial velocity is 10 m/s and the angle of projection with the inclined plane is 30°. The magnitude of velocity at the striking point P is



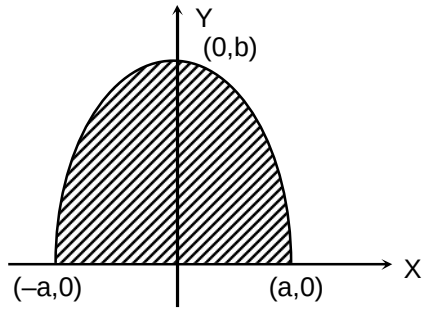
   (A) $5\ ms^{-1}$
   (B) $5\sqrt{3}\ ms^{-1}$
   (C) $\frac{10}{\sqrt{3}} ms^{-1}$
   (D) $\frac{10}{3} ms^{-1}$

3. A man pulls a heavy block of mass m on a rough horizontal track starting from rest. Force F applied by the man is always horizontal. Coefficient of friction between the block and the ground is given by $\mu = \mu_0 - \lambda v$, until it becomes zero. Where μ_0 and λ are positive constants and v is the speed of the block. Acceleration of the block when $v > \frac{\mu_0}{\lambda}$ is
   (A) $\frac{F}{m}$
   (B) $\frac{F}{m} - g\mu_0 \ln v$
   (C) $\frac{F}{m} - \mu_0 g$
   (D) $(\mu_0 - \lambda_v)g$

4. Potential energy of a particle moving in x-y plane as a function of co-ordinates is given by $U(x,y) = \frac{y^2}{2x}$. If particle is constrained to move such that its speed remains constant and no other force except conservative force acts on the particle then its path should be
   (A) Circle with centre at origin
   (B) Parabola

(C) Ellipse
(D) Can't be determined

5. Coordinates of the centre of mass of a lamina shaped as semi-ellipse (assuming uniform distribution of mass) as shown in the figure are



(A) $\left(0, \frac{4a}{3\pi}\right)$

(B) $\left(0, \frac{4b}{3\pi}\right)$

(C) $\left(0, \frac{2(a+b)}{3\pi}\right)$

(D) $\left(0, \frac{2b}{\pi}\right)$

6. A massless cylindrical container filled with an ideal liquid is placed on a smooth horizontal surface. A small orifice of cross-section area a is made in the side wall of the container near the bottom of container. If time taken by the liquid to completely flush out of the container is t, then velocity of the container after a long time is (Assuming area of cross section of container A >> a).

(A) $\frac{2a}{A}.gt$

(B) gt

(C) $\frac{A}{a}.gt$

(D) $\frac{a}{A}.gt$

7. A particle executing S.H.M. along a straight line has time period 6 seconds. Its displacement from its mean position is half of its amplitude at certain time and it is moving away from mean position. Time taken by the particle to cross the mean position for the first time is

(A) 5 sec
(B) 1 sec
(C) $\frac{5}{2}$ sec
(D) 3 sec

8. An observer moving with a speed v towards a stationary source records a frequency f. The value of frequency recorded by the same observer if he recedes with a velocity v/2 from the same source is (c = speed of sound in medium)

(A) $f\left(\frac{c-v}{c+v}\right)$

(B) $f\left(\frac{2c-v}{c+v}\right)$

(C) $\frac{f}{2}\left(\frac{2c-v}{c+v}\right)$

(D) $\frac{f}{2}\left(\frac{c-v}{2c+v}\right)$

9. Two planets A and B revolve around the same sun in different circular orbits. If $m_A : m_B = 1 : 2$ and $T_A : T_B = 8 : 1$. Where m_A, m_B are the masses and T_A, T_B are the time periods of planets A and B respectively. The ratio of angular momentum of planet A and planet B about their common sun is

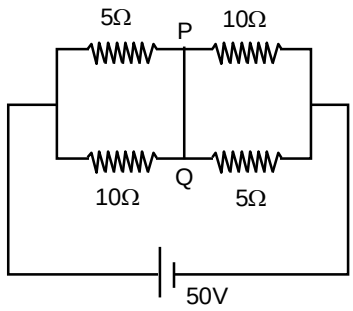
(A) $\sqrt{2}$
(B) 1
(C) 4
(D) 8

10. Magnitude of electric field at a point on the axis of a uniformly charged ring is E. If E_1 be the magnitude of electric field due to half portion of same ring at the same point on the axis, then

(A) $E_1 = E/2$
(B) $E_1 > E/2$
(C) $E_1 < E/2$
(D) $E_1 > E$

11. In the circuit shown in the figure. The value of current in the wire connected between P and Q is

(A) 5 A
(B) 10 A
(C) 15 A
(D) 2.5 A



12. A wire of length ℓ is used to form an n-sided regular polygon. Wire carries a constant current i. Let B_n be the magnitude of magnetic field at the centre of regular polygon, then B_n is given by

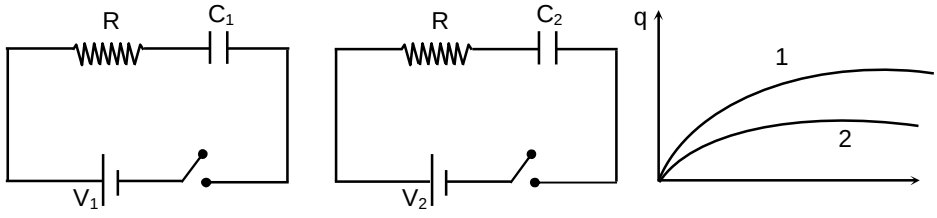
(A) $\frac{\mu_0 ni}{\pi\ell}\tan\left(\frac{\pi}{n}\right)$

(B) $\frac{\mu_0 i}{\pi\ell}$

(C) $\frac{\mu_0 n^2 i}{\pi\ell}\frac{\sin^2\left(\frac{\pi}{n}\right)}{\cos\left(\frac{\pi}{n}\right)}$

(D) $\frac{\mu_0 in^2}{\pi\ell}\cos\left(\frac{\pi}{n}\right)$

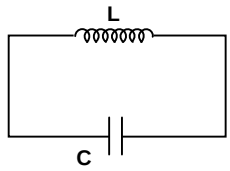
13. The graph of q versus t is plotted for two different RC series circuits. Both circuits have same resistance but different values of capacitance and source battery as shown in the figure.
Pick the correct option:



Circuit-1 Circuit-2

(A) $V_1 < V_2$ if $C_1 = C_2$
(B) $V_1 > V_2$ if $C_1 > C_2$
(C) $V_1 = V_2$ if $C_1 = C_2$
(D) $V_1 > V_2$ if $C_1 < C_2$

14. In L-C circuit shown in the figure, the value of current at t = 0 is i_0 and charge on the capacitor at the same moment is $i_0\sqrt{3LC}$. The value of the maximum current through the inductor is



(A) $2i_0$
(B) i_0
(C) $\sqrt{2}\,i_0$
(D) $\sqrt{5}\,i_0$

15. Electric field associated with an electromagnetic wave in vacuum is given by $\vec{E} = E_0 \sin(\omega t - Kz)\hat{i}$. The value of magnetic field associated with the same wave is (c is the speed of light in vacuum)

(A) $\vec{B} = E_0 c \sin(\omega t - Kz)\hat{j}$
(B) $\vec{B} = \frac{E_0}{c}\sin(\omega t + Kz)\hat{i}$
(C) $\vec{B} = \frac{E_0}{c}\sin(\omega t - Kz)\hat{j}$
(D) $\vec{B} = \frac{E_0}{c}\sin(\omega t - Ky)\hat{k}$

16. Light emitting diode is constructed using p-n junction (E_g = 1.5 eV). What is the wavelength of light emitted by this LED.

(A) 720 nm
(B) 560 nm
(C) 450 nm
(D) 827 nm

17. Fractional error in density of a sphere of radius r = (2$\pm$ 0.04) cm is

(A) 0.06
(B) 0.12
(C) 0.04
(D) 0.08

18. A beam of monochromatic light is incident on a slit of width 40 μm perpendicularly. If wavelength of light is 400 nm. The value of angular width of central maxima is

(A) 0.04 rad
(B) 0.002 rad
(C) 0.02 rad
(D) 0.01 rad

19. A particle is projected with a velocity $\vec{V} = V_x\hat{i} + V_y\hat{j}$ in a region having magnetic field $\vec{B} = B_0\hat{i}$ and electric field $\vec{E} = E_0\hat{j}$. If charge on the particle is q and mass m, then the value of acceleration of the particle just after the projection is

(A) $\frac{q}{m}\left(E_0\hat{j} - V_y B_0\hat{k}\right)$

(B) $\frac{q}{m}\left(E_0\hat{j} + V_yB_0\hat{k}\right)$

(C) $\frac{q}{m}\left(E_0\hat{j} - V_yB_0\hat{i}\right)$

(D) $\frac{q}{m}\left(E_0\hat{j} + V_xB_0\hat{i}\right)$

20. An X-Ray tube operates at potential difference 15 kV. The value of wavelength which is absent in the X-Ray spectrum generated is
(A) 5 nm
(B) 0.05 nm
(C) 90 pF
(D) 8 Å

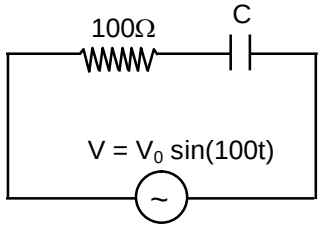
SECTION – B
(Numerical Answer Type)

This section contains **10** questions. The answer to each question is a **NUMERICAL VALUE.** For each question, enter the correct numerical value (in decimal notation, truncated/rounded-off to the **second decimal place**; e.g. XXXXX.XX).

21. A carnot engine working on two temperature reservoirs takes 800 J heat per second from high temperature reservoir of temperature 400 K. Let Q be the heat exhaust per second to the low temperature reservoir of temperature T. Then the value of $\frac{Q}{T}$ (in J/S-K) is

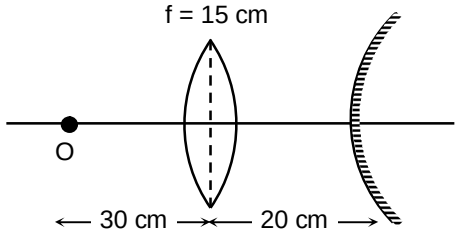
22. A uniform disc of mass m is moving on an inclined surface without slipping. Let K_1 be the kinetic energy of the disc with respect to center of mass of the disc while K_2 be the kinetic energy of disc with respect to a point present on the disc and moving with the fastest speed, then the value of $\frac{K_2}{K_1}$ is

23. A series R-C circuit is connected to an AC supply as shown in the figure. If power factor of the circuit is $\frac{1}{\sqrt{5}}$, then the value of the capacitance C (in μF) is



24. Consider a nuclear reaction
$A^{200} \rightarrow B^{120} + C^{80}$
If binding energy per nucleon of A, B and C are 5.5 MeV, 6 MeV and 6.5 MeV respectively. Find the Q-value (in MeV) of the reaction.

25. Image of an object shown in the figure is formed at the object itself. The value of the radius of curvature (in cm) of the convex mirror is

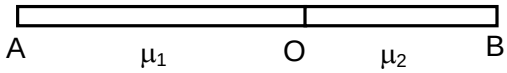


26. An ac voltage source $E = 200\sqrt{2}\sin(100t)$ volt is connected across a circuit containing an ac ammeter and a capacitor of capacitance 1mF. The reading of the ammeter (in ampere) is

27. A car moves at a constant speed of 36 km/h on a level road. The coefficient of friction between the tyre and the road is 0.8. The car negotiates a curve of radius R. If g = 10 ms^{-2}, the minimum value of R (in meter) for which the car will move without sliding is

28. A plano-convex lens, when silvered on the plane side, behaves like a concave mirror of focal length 30 cm. When it is silvered on the convex side, it behaves like a concave mirror of focal length 10 cm. The refractive index of the material of the lens is

29. Two strings of mass per unit length μ_1 = 0.09 kg/m and μ_2 = 0.01 kg/m are joined together and stretched between two supports as shown in the figure. A sinusoidal wave of amplitude 3.5 cm is sent from the end A. Find the amplitude (in cm) of the reflected wave from the joint 'O',



30. The equation of motion of a bike in a straight line is given by $x = 4t + 3t^2 - t^3$, where x is in meter and t is in seconds. The maximum velocity (in m/s) of the bike is

Chemistry

PART – B

SECTION – A
(One Options Correct Type)

This section contains **20 multiple choice questions**. Each question has **four choices** (A), (B), (C) and (D), out of which **ONLY ONE** option is correct.

31. Which of the following metal is present in the ore cerussite?
(A) Fe
(B) Pb
(C) Zn
(D) Al

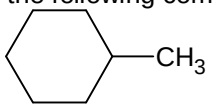
32. The pairs of enantiomers possible for the complex $[Ma_2b_2cd]$ is/are
(A) 1
(B) 3
(C) 2
(D) 0

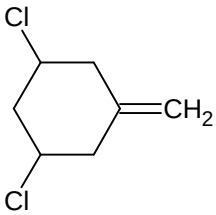
33. The correct match between Item-I and Item-II

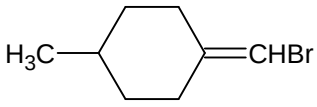
	Item – I		Item – II
(a)	Terfenadine	(I)	Tranquilizers
(b)	Chlordiazepoxide	(II)	Analgesics
(c)	Morphine	(III)	Antihistamines
(d)	Chloramphenicol	(IV)	Antibiotic

(A) (a – I), (b – III), (c – IV), (d – II)
(B) (a – III), (b – I), (c – II), (d – IV)
(C) (a – II), (b – III), (c – I), (d – IV)
(D) (a – IV), (b – II), (c – III), (d – I)

34. Nitrogen, sulphur, halogens, phosphorous present in an organic compound are detected by Lassaigne test. The formation of prussian blue colour confirms the presence of nitrogen. The compound which is blue in colour is
(A) $\left[Fe(CN)_5NOS\right]^{4-}$
(B) $\left[Fe(SCN)\right]^{2+}$
(C) $Fe_4\left[Fe(CN)_6\right]_3.xH_2O$
(D) NaCN

35. Which of the following compounds exhibit geometrical isomerism?
(A) 

(B) 

(C) 

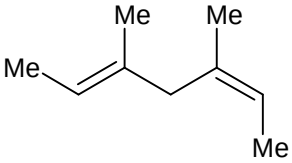
(D) $CH_2 = C = CH_2$

36. The correct order of third ionization energy for dipositive ion is
(A) $Be^{2+} > N^{2+} > B^{2+} > C^{2+}$
(B) $N^{2+} > B^{2+} > C^{2+} > Be^{2+}$
(C) $Be^{2+} > C^{2+} > N^{2+} > B^{2+}$
(D) $C^{2+} > N^{2+} > B^{2+} > Be^{2+}$

37. The H_3PO_3 formed on reaction of 3.6 g acetic acid with PCl_3 can be neutralized by
(A) 200 ml of 0.02 M $Ba(OH)_2$
(B) 200 ml of 0.1 M $Ba(OH)_2$
(C) 200 ml of 0.3 M NaOH
(D) 200 ml of 0.1 M $Al(OH)_3$

38. When acetamide is heated with anhydrous phosphorous pentoxide forms A. A on further treatment with CH_3MgBr followed by hydrolysis produces B. Which of the following statement regarding B is correct?
(A) B on reaction with hydroxylamine produces diastereomers
(B) Schiff test is used for detection of compound B
(C) B on reaction with iodine in presence of base gives yellow precipitate
(D) B forms silver mirror on treatment with Tollens reagent

39. Catalytic hydrogenation of the following compound produces saturated hydrocarbons. The number of stereoisomer(s) formed is

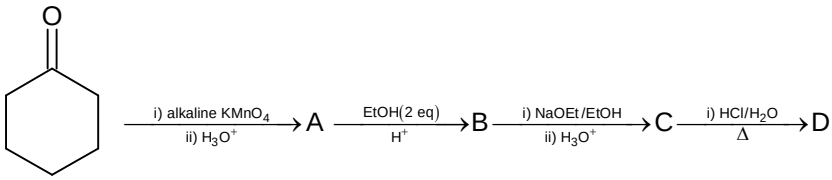


(A) 1
(B) 2
(C) 3
(D) 4

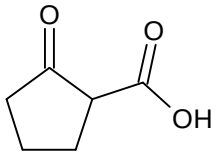
40. In the following Latimer diagram, the species undergoes disproportionation reaction is

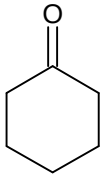
$$MnO_4^- \xrightarrow{E^o=0.56} MnO_4^{2-} \xrightarrow{E^o=0.27} MnO_4^{3-} \xrightarrow{E^o=0.93} MnO_2 \xrightarrow{E^o=0.15} Mn_2O_3 \xrightarrow{E^o=-0.23} Mn(OH)_2 \xrightarrow{E^o=-1.56} Mn$$

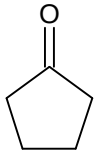
(A) MnO_4^{2-}
(B) MnO_4^{3-}
(C) Mn_2O_3
(D) $Mn(OH)_2$

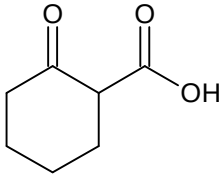
41. 

The product (D) is

(A) 

(B) 

(C) 

(D) 

42. Consider the equilibrium, $A(g) \rightleftharpoons B(g) + C(g)$

At a constant pressure of 1 atm, A dissociated to the extent of 50% at 500 K. Calculate the $\Delta G^{o}\left(kJ\ mol^{-1}\right)$ for the reaction

(A) 4.61
(B) 4608
(C) 6540
(D) 6.54

43. The total number of lone pairs in SF_4 is

(A) 1
(B) 2
(C) 12
(D) 13

44. $C_6H_5 - C \equiv CH \xrightarrow[HgSO_4]{H_2SO_4} P \xrightarrow[(2)\ H^+,\Delta]{(1)\ PhCHO,OH^-} Q \xrightarrow[\Delta]{NH_2-NH_2/OH^-} R \xrightarrow[(2)\ Zn/H_2O]{(1)\ O_3} S + PhCHO$.

The compound (S) is

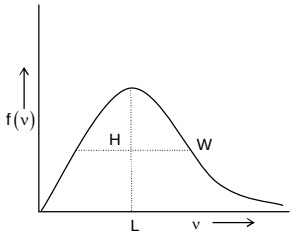
(A) $C_6H_5 - \overset{\overset{O}{\|}}{C} - CH_3$

(B) $C_6H_5 - \overset{\overset{O}{\|}}{C} - CH = CH - Ph$

(C) $C_6H_5CH_2CH = CHPh$

(D) $C_6H_5CH_2CHO$

45. The Maxwell Boltzmann distribution for molecular speed is shown in the following figure:



In the figure, H is height of peak, L is location of the maximum and W is width of half height, if the temperature is decreased,

(A) H increases, L decreases and W increases
(B) H increases, L decreases and W decreases
(C) H decreases, L decreases and W increases
(D) H decreases, L decreases and W decreases

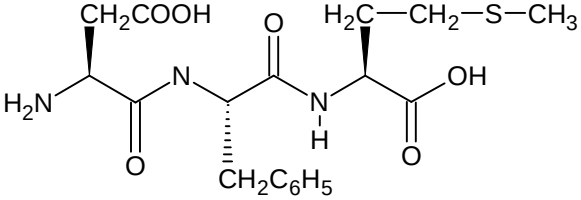
46. The atomicity and the total number of bonds in the elemental white phosphorous molecule are respectively

(A) 4 and 6
(B) 6 and 4
(C) 4 and 4
(D) 6 and 6

47. The electron of a hydrogen atom is in its nth Bohr orbit having de-Broglie wavelength of $13.4\,\mathring{A}$. The value of n is

(A) 2
(B) 4
(C) 6
(D) 5

48. The correct order of aminoacids present in the tripeptide given below is



(A) Glu – Phe – Met
(B) Asp – Trp – Cys
(C) Asp – Phe – Met
(D) Glu – Phe – Cys

49. Which of the following statement is INCORRECT?

(A) Be and Al ions have strong tendency to form complexes.
(B) Li_2O and MgO combine with excess oxygen to give superoxide.
(C) Lithium hydrogencarbonate is not obtained in solid form while all other elements form solid hydrogencarbonates.
(D) Diborane is produced by the reaction of BF_3 with sodium hydride.

50. Which of the following water is best suitable for drinking purpose?

(A) The water with nitrate concentration 200 ppm.
(B) The water with Fe concentration 10 ppm.
(C) The water with Zn concentration 3 ppm.
(D) The water with sulphate concentration 600 ppm.

SECTION – B
(Numerical Answer Type)

This section contains **10** questions. The answer to each question is a **NUMERICAL VALUE.** For each question, enter the correct numerical value (in decimal notation, truncated/rounded-off to the **second decimal place**; e.g. XXXXX.XX).

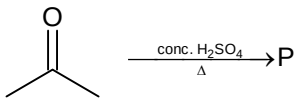
51. The ratio of number of chiral centres and number of carbon atoms in terpineol is

52. The mass of $Ca(NO_3)_2$ in mg that is present in 50 ml of solution with 9.50 ppm of Ca is……
$\left(\text{Molar mass of } Ca(NO_3)_2 = 164 \text{ g mol}^{-1}\right)$

53. The pressure exerted by 10^{23} particles of gas each of mass 10^{-22} g in a container of volume 1 dm^3 is $x \times 10^5$ Pa. The root mean square speed is 10^5 cms^{-1}. Find the value of x.

54. Consider an isothermal reversible compression of one mole of an ideal gas in which the pressure of system is increased from 5 atm to 30 atm at 300 K. The entropy change of surrounding in $\left(JK^{-1}\right)$ is…..

55. In 200 g of water, 0.01 mole of NaCl and 0.02 mole of sucrose are dissolved. Assuming solution to be ideal, the depression in freezing point of water (in ^oC) will be
…….$\left(K_f \text{ of water} = 1.86^oC/m\right)$

56. 

The above reaction leads to formation of product P. The mass percentage of carbon in P is……

57. The number of molecules with energy greater than threshold energy for a reaction increases ten fold by a rise of temperature from 37^oC to 67^oC. Its energy of activation in J/mol is…..

58. An element with molar mass 4.5×10^{-2} kg mol^{-1} forms cubic unit cell with edge length 322 pm. If its density is 4.5×10^3 kg m^{-3}, the radius of element is approximately…… $\times 10^{-12}$ m.

59. 0.85 g of ammonia is mixed in 250 ml of 0.1 M NaOH. This mixture is now diluted to 500 ml. 20 ml of this solution is now taken in another container. $\frac{1}{2}$ml of 5M HCl is added to this. Find pH of this solution $\left(pK_b = 4.74\right)$

60. PCl_5 in solid state exist as oppositely charged ions
$$\frac{\text{Number of Cl in anion of } PCl_5 \text{ in solid state}}{\text{Number of Cl in cation of } PCl_5 \text{ in solid state}} = x$$
Find the value of x.

Mathematics

PART – C

SECTION – A
(One Options Correct Type)

This section contains **20 multiple choice questions**. Each question has **four choices** (A), (B), (C) and (D), out of which **ONLY ONE** option is correct.

61. If $f(n)=\sum_{r=0}^{n}\frac{x_r^3}{1-3x_r+3x_r^2}$ for $x_r=\frac{r}{n}$, $\forall\ r \in \{0, 1, 2, ... n\}$, then the value of f(101) will be

(A) 1
(B) 101
(C) 51
(D) 100

62. If $P(r)=\int_0^{\pi/2}\frac{dx}{1+(\tan x)^r}$, then the value of $P\left(\sqrt{2}\right)+P(0)+P\left(-\sqrt{2}\right)$ is

(A) $\frac{3\pi}{2}$

(B) $\frac{3\pi}{4}$

(C) $\frac{\pi}{2}+\frac{1}{2}$

(D) $\pi+\frac{1}{2}$

63. A die is tossed. If the die lands on '1' or '2', then one coin is tossed. If the die lands on '3', then two coins are tossed. Otherwise, three coins are tossed. Given that the resulting coin tosses produced no 'heads', what is the probability that the die landed on '1' or '2'

(A) $\frac{13}{48}$

(B) $\frac{1}{6}$

(C) $\frac{1}{2}$

(D) $\frac{8}{13}$

64. Suppose that the roots of P(x) = 0, where $P(x) = x^3 - 2kx^2 - 3x^2 + hx - 4$ are distinct, and that P(k) = P(k + 1) = 0, then the of value of 'h' is

(A) 8
(B) 2
(C) –4
(D) –2

65. If $|ax^2 + bx + c| \le 1$ for all $x \in [-1, 1]$, then

(A) $|a + c| > 1$
(B) $|c| > 1$
(C) $|c| > 2$
(D) $|a + c| \le 1$

66. Let a and b are real numbers such that $a^2 + 2\cos b = 1$, then number of integers in the range of $(a - \cos b)$ will be
(A) 0
(B) 2
(C) 4
(D) 1

67. In how many ways, can three collinear points be selected from 16 integer points (a, b), $a, b \in Z$, such that $a, b \in \{1, 2, 3, 4\}$
(A) 40
(B) 38
(C) 44
(D) 52

68. The exhaustive set of domain of the function $f(x) = \sqrt{5 - 3(x-2)^2 - 4\sqrt{\frac{1}{4}x^2 - x + 1}}$ is
(A) $\left[\frac{1}{3}, 3\right]$
(B) [1, 3]
(C) [–1, 3]
(D) $\left[-3, -\frac{1}{3}\right]$

69. How many complex numbers ($z = x + iy$, $x, y \in I$ (integer), $i = \sqrt{-1}$), will satisfy $|z|^2 - 7|z| + 10 \le 0$ and $\left|\frac{z+3}{z-3}\right| \le 1$, simultaneously
(A) 28
(B) 34
(C) 36
(D) 40

70. If $A = \lim\limits_{x\to\infty} e^{-\sec\left(\tan^{-1}x\right)}$ and $B = \lim\limits_{x\to-\infty} e^{\tan\left(\sec^{-1}x\right)}$, the value of $|A| + |B|$ is
(A) greater than $\frac{\pi}{2}$
(B) less than 1
(C) equal to 2
(D) equal to $\frac{\pi}{2}$

71. If $\frac{x^2}{4} + \frac{y^2}{1} = 1$ is an ellipse, centre at O(0, 0) and A, B, C are three points on the same ellipse, such that, segment OA, OB, OC make angles $\frac{\pi}{6}, \frac{2\pi}{3}, \frac{5\pi}{6}$ respectively, with the positive x-axis, then the value of $\frac{1}{(OA)^2} + \frac{1}{(OB)^2} + \frac{1}{(OC)^2}$ is
(A) $\frac{27}{16}$

(B) $\frac{51}{4}$
(C) 3
(D) 4

72. The number of focal chord(s) of the parabola $y^2 = 8x$ which are tangent to the circle $(x-4)^2 + (y-2)^2 = 4$ is/are
(A) 1
(B) 2
(C) 3
(D) 0

73. If set A is the range of the function $y = [\sin^{-1} x]$, (where [.] denotes greatest integer function) and B is the set of integers in the range of $f(x) = \frac{x^2 - 3x - 4}{x^2 - 3x + 4}$, then $n(A \cap B)$ is
(A) 4
(B) 1
(C) 0
(D) 3

74. The number of distinct value(s) of x, where $x \in [0, 1]$ which satisfies the equation $\int_0^x \frac{t^2+1}{t^4+t^2+1} dt = \cos^{-1}\left(\cos\left(\sqrt{9x^2 - 12x + 4}\right)\right)$ is
(A) 1
(B) 0
(C) 2
(D) infinite

75. If $P = \begin{bmatrix} 0 & \cos A & \cos A \\ 2\sin B & \sin B & -\sin B \\ \tan C & -\tan C & \tan C \end{bmatrix}$ is a matrix, such that $PP^T = I$, (I is identity matrix), then the value of $\cos^2 A + \cos^2 B + \cos^2 C$ is
(A) 3
(B) 1
(C) $\frac{11}{12}$
(D) $\frac{25}{12}$

76. In ΔABC, let a, b, c be the length of the sides opposite to angles A, B, C respectively, and $\frac{b+c-a}{4} = \frac{a+c-b}{3} = \frac{a+b-c}{2}$. If inradius is $\frac{2\sqrt{2}}{\sqrt{3}}$ units, then the circumradius of the triangle formed by joining the feet of the three altitude of the triangle A, B, C, is
(A) $\frac{35}{4\sqrt{6}}$
(B) $\frac{35}{\sqrt{6}}$

(C) $\frac{35}{2\sqrt{6}}$

(D) $\frac{35}{8\sqrt{6}}$

77. If V is the volume of the cuboid whose lengths of edges are a, b, c such that a > b > c, and a, b, c $\in$ N and a, b, c satisfies the equation $\frac{1+bc}{37-ab}=\frac{1}{a}$, then the value of $V_{max} - V_{min}$ is

(A) 3
(B) 2
(C) 4
(D) 10

78. Let $\vec{a}, \vec{b}, \vec{c}$ be three distinct unit vectors and $P_1 = \max$ of $\left(|\vec{a}-\vec{b}|+|\vec{b}-\vec{c}|+|\vec{c}-\vec{a}|\right)$ and $P_2 = \max$ of $\left(|\vec{a}-\vec{b}|^2+|\vec{b}-\vec{c}|^2+|\vec{c}-\vec{a}|^2\right)$, then $\frac{P_1}{P_2}$ is equal to

(A) $\frac{1}{\sqrt{3}}$

(B) $\frac{1}{\sqrt{2}}$

(C) 1

(D) $\sqrt{2}$

79. The top of a hill observed from the top and the bottom of a building of height 10 meters is at angles of α and β respectively, then the height of the hill (in meters) is

(A) $\frac{10\cos\alpha\sin\beta}{\sin(\beta-\alpha)}$

(B) $\frac{10\sin\alpha\sin\beta}{\sin(\beta-\alpha)}$

(C) $\frac{10\cos\alpha\cos\beta}{\sin(\beta-\alpha)}$

(D) none of these

80. The area of the region bounded by the curve $y=\sqrt{5-x^2}$ and y = |x – 1| is

(A) $\frac{1}{2}\left(\sin^{-1}\frac{1}{2}-\sin^{-1}\frac{3}{5}\right)$

(B) $\frac{5\pi}{4}-1$

(C) $\frac{5\pi}{4}-\frac{1}{2}$

(D) $\frac{1}{2}\left(\sin^{-1}\frac{1}{\sqrt{5}}+\sin^{-1}\frac{3}{5}\right)$

SECTION – B
(Numerical Answer Type)

This section contains **10** questions. The answer to each question is a **NUMERICAL VALUE.** For each question, enter the correct numerical value (in decimal notation, truncated/rounded-off to the **second decimal place**; e.g. XXXXX.XX).

81. If the locus of the midpoint of portion of tangents intercepted between coordinate axes of the hyperbola $\frac{x^2}{16}-\frac{y^2}{9}=1$, is $\frac{\alpha}{x^2}-\frac{9}{y^2}=\beta$, then $(\alpha+\beta)$ is equal to

82. If ${}^{31}C_5-{}^{30}C_4-{}^{29}C_4-{}^{28}C_4-.....-{}^{k+1}C_4-{}^{k}C_4=1$, then the value of k^2 is

83. Let $k_1, k_2, k_3, ..., k_n$ are in Arithmetic Progression. If $\sum_{r=1}^{100}\frac{1}{k_r k_{r+1}}=10$ and $k_2 + k_{100} = 50$, then the value of $\frac{(k_{101}-k_1)^2}{10}$ is

84. If chords of the curve $4x^2 + y^2 – x + 4y = 0$ which subtends $90°$ at (0, 0), passes through a fixed point (α, β), then $|\alpha + \beta|$ is

85. If p, q, r are the lengths of the sides of triangle PQR and $\frac{p-2}{r-q}=1$ and $2pq = r^2 + 4$ and if R_1 is the circum radius of ΔPQR, then $(R_1)^2$ is equal to

86. In triangle ABC, if $a\cos A + b\cos B + c\cos C = S$, where S is semiperimeter, and if $p=\frac{a\sin A+b\sin B+c\sin C}{S}$, then the value of $(10p)^2$ is

87. Let $M=\begin{bmatrix}1 & 2 & 0\\ 0 & 2 & 0\\ 0 & -1 & 2\end{bmatrix}$ be a matrix and $A – 8M = M^4$ and $B + 5M^3 = 8M^2 – 10M$, then the value of $\left(\frac{\det(A+B)}{\text{tr}(A+B)}\right)^2$ is

88. If $P=\lim_{n\to\infty}\sum_{r=1}^{n}\frac{r^3}{n^5}$ and $Q=\lim_{n\to\infty}\sum_{r=1}^{n}\frac{r^7}{n^8}$, then the value of Q – P is

89. If α, β, γ are real roots of the equation $x^3 – 3x^2 + (a + 2)x – a = 0$ and $\alpha < \beta < \gamma$, $a \in R$, then the value of $(4\alpha – \alpha^2 + \gamma^2)^3$ is equal to

90. In the complex plane, let A be the set of solutions to $z^3 = 8$ and let B be the set of solutions to $z^3 – 8z^2 – 8z + 64 = 0$. If L is the greatest distance between a point of set A and a point of set B, then the value of L^2 is